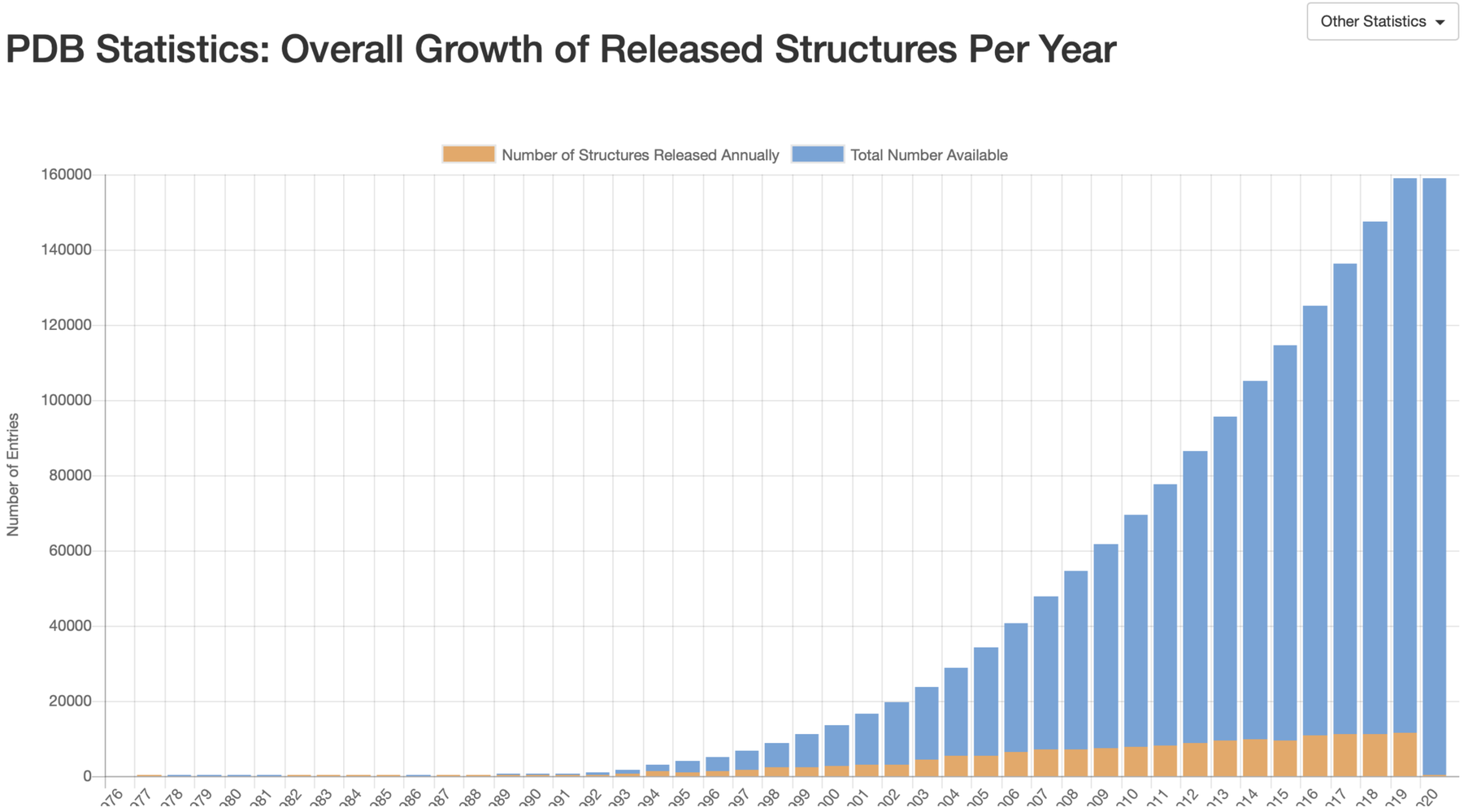
Chem 456 1/28/2026 Target Selection

- This lecture is intended to help you achieve the following learning objective: Analyze the prospects of a biological macromolecule (usually a protein) as a target for ligand design, from a community, scientific, and business perspective.
- This module will consist of a lecture interspersed with discussions, an interactive database search exercise, and a review
- At the end of this module, you should be able to address the question of what is necessary and what is desirable for a biological macromolecule to be a suitable target for structure-based drug design.
  - What is necessary/desirable about the physiological role?
  - What is necessary/desirable about the structural properties?
  - What are some business considerations that pharmaceutical companies use to decide whether to develop a drug for a particular target?
- You should also know about some databases that you can use (possibly to follow up on a conversation with a large language model) to help find and investigate a suitable target for SBDD

The exponential growth of biomacromolecule structures is an opportunity for SBDD



https://www.rcsb.org/stats/growth/overall

Predictive structural modeling further expands the range of accessible targets

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Article | Open Access | Published: 15 July 2021

Highly accurate protein structure prediction with AlphaFold

John Jumper, Richard Evans, ... Demis Hassabis + Show authors

Nature **596**, 583–589 (2021) | Cite this article

566k Accesses | **983** Citations | **2993** Altmetric | Metrics

Abstract

Proteins are essential to life, and understanding their structure can facilitate a mechanistic understanding of their function. Through an enormous experimental effort[1,2,3,4], the structures of around 100,000 unique proteins have been determined[5], but this represents a small fraction of the billions of known protein sequences[6,7]. Structural coverage is bottlenecked by the months to years of painstaking effort required to determine a single

Deep learning takes on protein folding

Abstract

Supplementary Material

References and Notes

RESEARCH ARTICLE | PROTEIN FOLDING

Accurate prediction of protein structures and interactions using a three-track neural network

MINKYUNG BAEK, FRANK DIMAIO, IVAN ANISHCHENKO, JUSTAS DAUPARAS, SERGEY OVCHINNIKOV, GYU RIE LEE, JUE WANG, QIAN CONG, LISA N. KINCH, [...] DAVID BAKER +23 authors Authors Info & Affiliations

SCIENCE • 19 Aug 2021 • Vol 373, Issue 6557 • pp. 871-876 • DOI: 10.1126/science.abj8754

19,728 17 GET ACCESS

Deep learning takes on protein folding

In 1972, Anfinsen won a Nobel prize for demonstrating a connection between a protein's amino acid sequence and its three-dimensional structure. Since 1994, scientists have competed in the biannual Critical Assessment of Structure Prediction (CASP) protein-folding challenge. Deep learning methods took center stage at CASP14, with DeepMind's Alphafold2 achieving remarkable accuracy. Baek *et al.* explored network architectures based on the DeepMind framework. They used a three-track network to process sequence, distance, and coordinate information simultaneously and achieved accuracies approaching those of DeepMind. The method, RoseTTA fold, can solve challenging x-ray crystallography and cryo–electron microscopy modeling problems and generate accurate models of protein-protein complexes. —VV

What is necessary/desirable about the physiological role of a SBDD target?

What is necessary/desirable about the physiological role of a SBDD target?

Disease Relevance. Altering activity (usually by activation/inhibition) should affect disease progression.

Brainstorm: How can disease relevance be predicted?

How can disease relevance be predicted?

- Mechanistic rationale, e.g.
  - the enzyme HIV protease has key role in viral life cycle. blocking it will slow viral reproduction.
  - the H+/K+-ATPase system is involved in stomach acid production. blocking it will reduce stomach acid and can be used to treat acid reflux.
- Mutations in a protein known to affect disease progression
  - in the clinic
  - in the laboratory
- Chemical probes affect disease progression
- Essential targets are better, e.g. NADH:ubiquinone oxidoreductase (NQR) is essential for *Vibrio cholerae* (common gut infection in developing tropical countries) and *Chlamydia trachomatis* (common STD) [1] but not *Pseudomonas aeruginosa* (common hospital infection)

Liang, P.; Rosas-Lemus, M.; Patel, D.; Fang, X.; Tuz, K.; Juárez, O. Dynamic Energy Dependency of Chlamydia Trachomatis on Host Cell Metabolism during Intracellular Growth: Role of Sodium-Based Energetics in Chlamydial ATP Generation. J. Biol. Chem. 2018, 293 (2), 510–522. https://doi.org/10.1074/jbc.M117.797209.

Brainstorm: What is necessary/desirable about the structural properties of a SBDD target?

What is necessary about the structural properties of a SBDD target?

- High-resolution structure available
  - Resolution of < 2.5 Å [1]
  - Homology models less trustworthy
    - can still be useful, perhaps to guide structure determination
  - Experimental structures/homology models can be refined by molecular dynamics simulation
- "Druggable"
  - precedence - previously targeted
  - structure-based
    - cavities or pockets
    - comparing physicochemical and geometric properties of pocket with known druggable targets
  - as a caveat, concept is about history and does not account for innovation

[1] Anderson, A. C. The Process of Structure-Based Drug Design. Chemistry & Biology 2003, 10 (9), 787–797. https://doi.org/10.1016/j.chembiol.2003.09.002.

What is desirable about the structural properties of a SBDD target?

- Small
  - less ambiguity regarding binding site
  - more amenable to molecular simulation
- Rigid
  - molecular docking is more accurate
  - molecular simulation requires less sampling
- Binding site is
  - unique, favoring specificity
  - evolutionarily conserved, less susceptible to resistance

Using Generative AI

- Large language models can be a powerful tool, but should be used carefully.
- Evaluate its claims by reviewing citations. They may be hallucinated.
- Ask follow-up questions and, if helpful, provide contrary information.
- Investigate targets more deeply by reviewing information from databases.

Is succinate dehydrogenase a good antibiotic drug target?

Copilot

Here's a clear, evidence-based answer grounded in the search results you provided:

Is succinate dehydrogenase (SDH) a good antibiotic drug target?

Short answer:

Probably not. While SDH is essential for bacterial energy metabolism, it is *not* considered an ideal antibiotic target because the enzyme is *highly conserved across bacteria, fungi, plants, and humans*, creating major selectivity and safety challenges.

Below is a breakdown of why.

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1. SDH is highly conserved across life, making selective inhibition difficult

Succinate dehydrogenase (Complex II of the electron transport chain) plays the same core bioenergetic role in bacteria, fungi, plants, and mammalian mitochondria. Evidence shows the enzyme is broadly conserved and structurally similar across phyla.

- SDH is described as a core mitochondrial enzyme present in both the TCA cycle and ETC in eukaryotes and in prokaryotic metabolism. tsijournals
- Many SDH inhibitors (SDHIs) used in agriculture have been shown to inhibit SDH in **multiple phyla, including humans**, not only fungi. mdpi

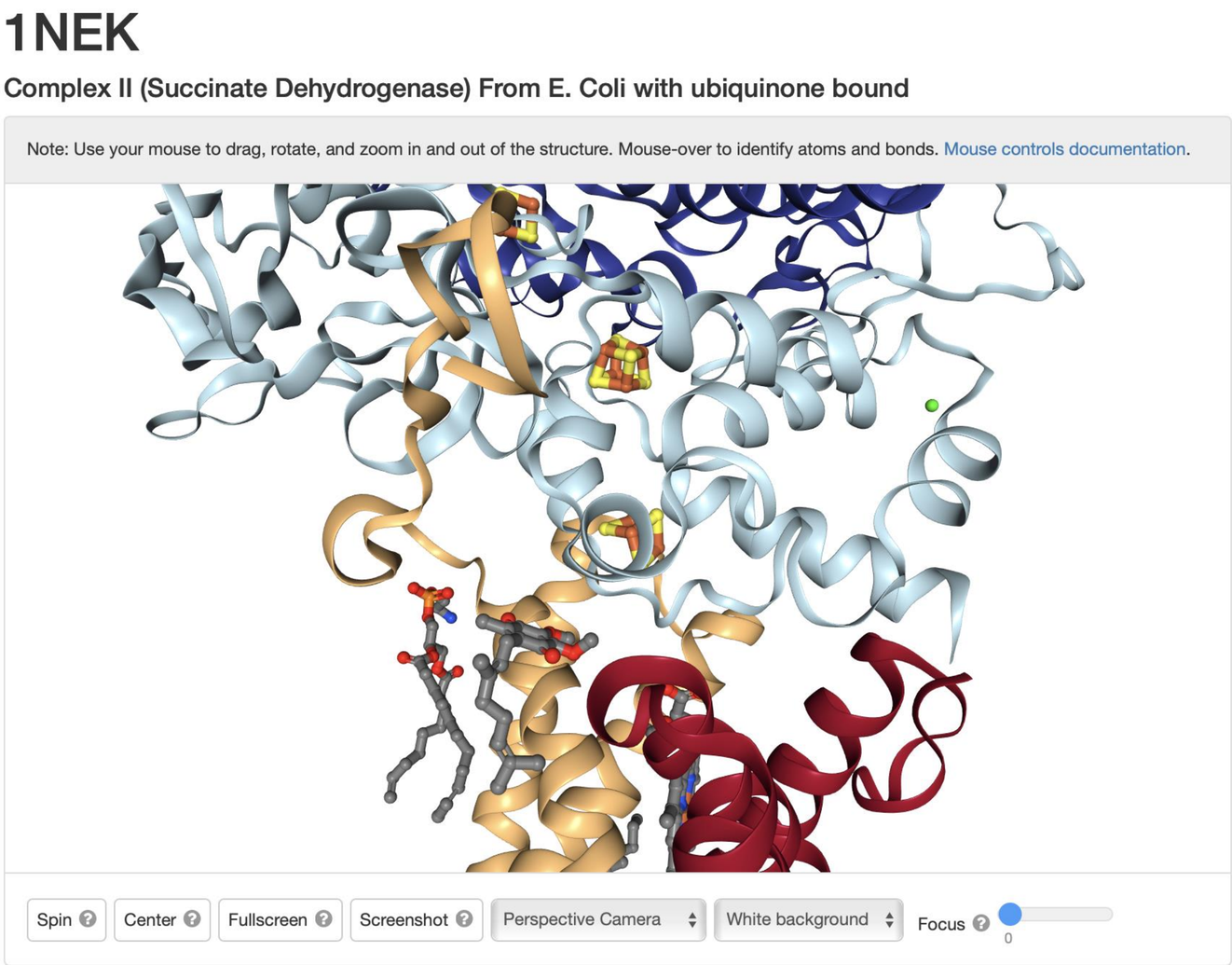
Implication: Designing inhibitors that affect bacteria but *not* human mitochondria is extremely challenging.

There are multiple online databases to help you investigate suitable targets

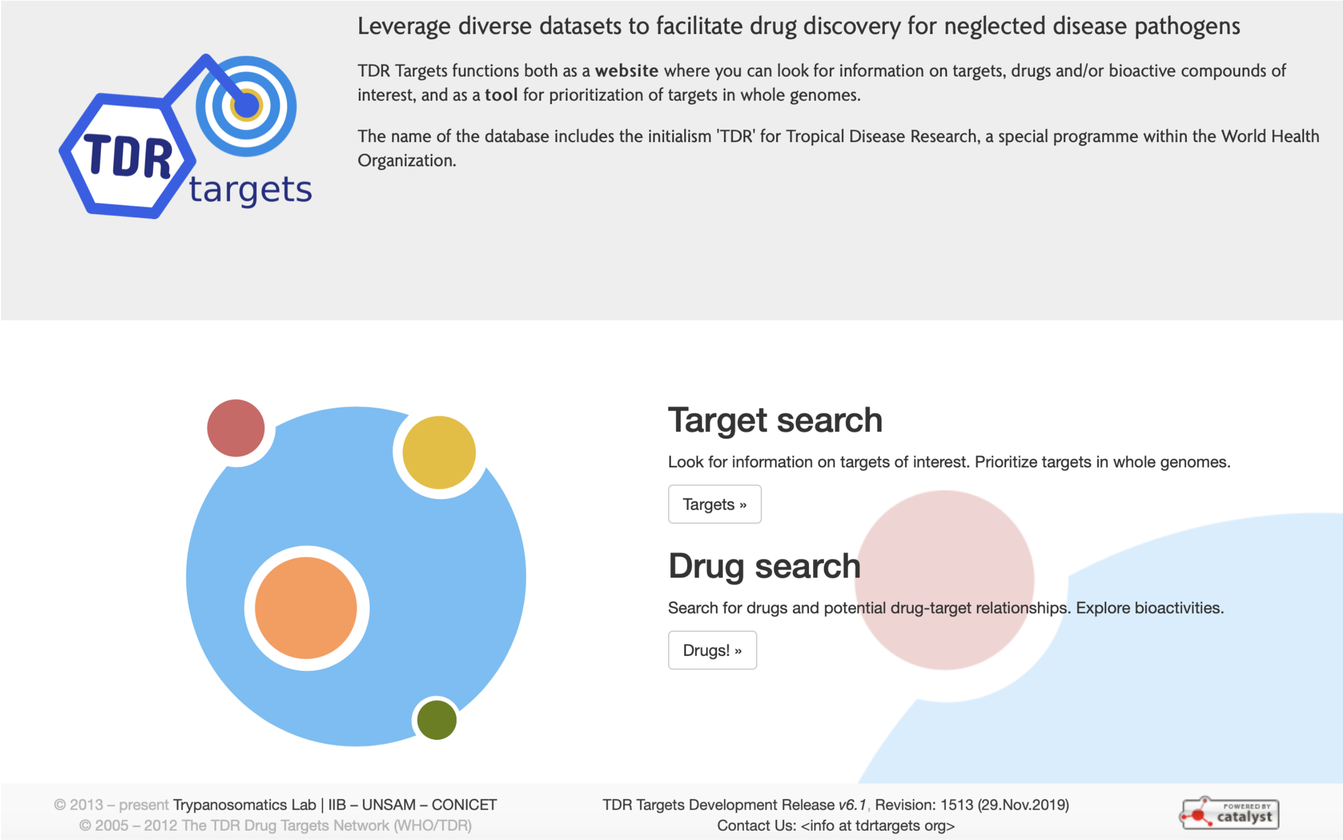
- The Special Programme for Research and Training in Tropical Diseases (TDR) Targets (https://tdrtargets.org/) - A database that allows you to apply various filters to identify drug targets. Focuses on neglected tropical diseases.
- Therapeutic target database (http://idrblab.net/ttd/) - “A database to provide information about the known and explored therapeutic protein and nucleic acid targets, the targeted disease, pathway information and the corresponding drugs directed at each of these targets.” This can help you determine precedence for a target.
- BindingDB (https://www.bindingdb.org/rwd/bind/index.jsp) - “BindingDB is a public, web-accessible database of measured binding affinities, focusing chiefly on the interactions of protein considered to be drug-targets with small, drug-like molecules.” BindingDB can help you determine whether a any small molecule (not necessarily a drug) has been found that binds to a target.

Interactive Database Exercise

- Oscar Juarez has worked on discovering inhibitors for Complex II, a.k.a. Succinate Dehydrogenase (SDH), from *Pseudomonas aeruginosa*
- The structure of E. Coli SDH has been solved at a resolution of 2.6 Å
- Subunit B (cyan) has Iron-Sulfur centers
- He has targeted the ubiquinone site at the interface of subunits B, C (yellow), and D (red)



- We think it's a good drug target but I want to see what the online databases suggest
- First, let's try TDR targets (https://tdrtargets.org). It is possible that an SDH drug for *P. Aeruginosa* may also work on some neglected tropical diseases.



- Try a “Quick Search…” for “succinate dehydrogenase”. This yields <u>many</u> results that are various subunits of the enzyme from different species
- Compared to the ubiquinone binding site, the “flavoprotein subunit” is on the opposite side of the complex

Target list

Search results for query: #2 (succinate dehydrogenase)

Show query parameters

Convert this list of targets into a list of drugs: More information?

Retrieve: All Associations (Curated and Predicted) | Curated Associations | Target Putative Associations (predicted)

97 records found | Showing page **1** of 4 (records 1-25) | Number of records to display 25 | Find orthologs in select species

Organism	Name	Ortholog group	Product
B. malayi	Bm1_17325	OG5_126927	succinate dehydrogenase [ubiquinone] flavoprotein subunit, mitochondrial
B. malayi	Bm1_17330	OG5_126927	succinate dehydrogenase [ubiquinone] flavoprotein subunit, mitochondrial
B. malayi	Bm1_17690	OG5_126893	succinate dehydrogenase [ubiquinone] iron-sulfur protein, mitochondrial
B. malayi	Bm1_30090	OG5_129614	Succinate dehydrogenase cytochrome b560 subunit, mitochondrialprecursor
B. malayi	Bm1_35660	OG5_129488	Succinate dehydrogenase
C. trachomatis	CT_591	OG5_126893	succinate dehydrogenase iron sulfur subunit
C. trachomatis	CT_592	OG5_126927	succinate dehydrogenase flavoprotein subunit
E. granulosus	EgrG_000416000	No group	succinate dehydrogenase ubiquinone iron sulfur

- A few of the hits had “iron-sulfur subunit” in the name. The iron-sulfur subunit is B, one of those next to the ubiquinone binding site.
- Try a “Quick Search…” for “succinate dehydrogenase iron-sulfur”.

Search results for query: #7 (succinate dehydrogenase iron-sulfur)

Show query parameters

Convert this list of targets into a list of drugs: More information?

Retrieve: All Associations (Curated and Predicted) | Curated Associations | Target Putative Associations (predicted)

9 records found | Showing page **1** of 1 (records 1-9) | Number of records to display 25 | Find orthologs in select species

Organism	Name	Ortholog group	Product
L. Loa (eye worm)	LOAG_10155	OG5_126893	succinate dehydrogenase iron-sulfur protein
M. ulcerans	MUL_1370	OG5_126893	succinate dehydrogenase iron-sulfur subunit
S. mansoni	Smp_089640.2	OG5_126893	succinate dehydrogenase iron-sulfur protein
S. mansoni	Smp_089640.3	OG5_126893	succinate dehydrogenase iron-sulfur protein
T. brucei	Tb927.8.3380	OG5_126893	succinate dehydrogenase iron-sulfur subunit
T. brucei	Tb927.9.5960	OG5_126893	succinate dehydrogenase iron-sulfur subunit, putative
T. cruzi	TcCLB.504949.30	OG5_126893	succinate dehydrogenase iron-sulfur subunit
T. cruzi	TcCLB.509769.60	OG5_126893	succinate dehydrogenase iron-sulfur subunit
W. endosymbiont of Brugia malayi	Wbm0600	OG5_126893	succinate dehydrogenase iron-sulfur subunit

- Click on Tb927.8.3380 and scroll down to Essentiality Data
- Suppose that we discover a SDH inhibitor in our SBDD campaign against the ubiquinone binding site of *P. Aeruginosa*. Based on these data, which other species would be most worth testing its efficacy against? Which would be least worth testing?

Essentiality

Tb927.8.3380 has direct evidence of essentiality

Gene/Ortholog	Organism	Phenotype	Source Study
mtu1581	Mycobacterium tuberculosis	non-essential	nmpdr
mtu3379	Mycobacterium tuberculosis	non-essential	nmpdr
Tb09.160.4380	Trypanosoma brucei	no significant loss or gain of fitness in bloodstream forms (3 days)	alsford
Tb09.160.4380	Trypanosoma brucei	significant gain of fitness in bloodstream forms (6 days)	alsford
Tb09.160.4380	Trypanosoma brucei	no significant loss or gain of fitness in procyclic forms	alsford
Tb09.160.4380	Trypanosoma brucei	significant gain of fitness in differentiation of procyclic to bloodstream forms	alsford
Tb927.8.3380 this record	Trypanosoma brucei	significant loss of fitness in bloodstream forms (3 days)	alsford
Tb927.8.3380 this record	Trypanosoma brucei	significant loss of fitness in bloodstream forms (6 days)	alsford
Tb927.8.3380 this record	Trypanosoma brucei	significant loss of fitness in procyclic forms	alsford
Tb927.8.3380 this record	Trypanosoma brucei	significant loss of fitness in differentiation of procyclic to bloodstream forms	alsford
b0724	Escherichia coli	non-essential	goodall
b4153	Escherichia coli	non-essential	goodall

- Also take a look at “Structural information”
- There is no crystal structures and the homology models from Modbase are not high quality. The homology model could be a guide, but I would not base an SBDD campaign against a *T. brucei* homology model.
- We could also look at AlphaFold

Structural information

Modbase 3D models:

There are 3 models calculated for this protein. More info on these models, including the models themselves is available at: Modbase

Target Beg	Target End	Template	Template Beg	Template End	Identity	Evalue	Model Score	MPQS	zDope
32	156	1zoy (B)	16	151	46.00	0	1	0.950829	0.33
32	156	4ysx (B)	40	175	46.00	0	0.99	0.956829	0.26
168	229	5i9f (A)	348	409	23.00	0.53	0.98	0.708498	-1.37

+ Help me make sense of these data.

Target Beg: first modeled residue
Target End: last modeled residue
Template: template structure used for modelling (PDB accession and chain)
Template Beg: first template residue in target-template alignment
Template End: last template residue in target-template alignment
Identity: sequence identity
Evalue: E value for target-template hit
Model Score: GA341 score (>0.7 for reliable model)
MPQS: ModPipe Quality Score (>1.1 for reliable model)

- Let's try "succinate dehydrogenase" in the Therapeutic Target Database (http://idrblab.net/ttd/) using "Search Target in Whole Database"
- Did you find SDH?

- If there are successful drugs for a target, they will be listed in a box ”Drugs and Modes of Action” (see below for MOR). Are there currently any drugs that target SDH?

Drugs and Modes of Action

List of Drug Status & Drug Name | List of MOA & Drug Name

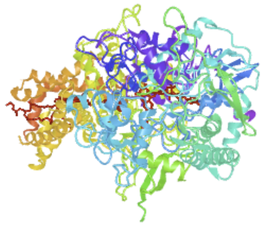
- Drug Status: **Approved Drug(s) (In total 24 Drugs)** *click to show more*
- Drug Status: **Clinical Trial Drug(s) (In total 16 Drugs)** *click to show more*
- Drug Status: **Discontinued Drug(s) (In total 15 Drugs)** *click to show more*
- Drug Status: **Preclinical Drug(s) (In total 2 Drugs)** *click to show more*

TTD **Therapeutic Target Database** POWERED BY: BIDD Bioinformatics and Drug Design group IDRB

Home | Target Search | Drug Search | Download | Help

Target Information

Target General Information (Target ID: T39811; Former ID: TTDI01376)

Target Name	Succinate dehydrogenase (SDHD)	
Synonyms	Succinate-ubiquinone reductase membrane anchor subunit; Succinat...	*click to show more/less*
Gene Name	SDHD SDH4	
Target Type	Literature-reported target	REF
Disease	0 Target-related Diseases	*click to show more/less*
Structure		
Options	PDB / 8GS8	
Structure	8GS8	
Method	EM	
Resolution	2.86 A	
Positions	D=1-159	
Sequence	MAVLWRLSAVCGALGGRALLLRTPVVRPAHISAFLQDRPIPEWCGVQHIH...	*click to show more/less*
Function	Membrane-anchoring subunit of succinate dehydrogenase (SDH) that ...	*click to show more/less*
Other Information	Family and Domain	*click to show more/less*
	External Links	*click to show more/less*

Target-Disease Associations based on Genetic Mutation | Target-Disease Associations based on Transcriptomic Analysis

- Finally let's try BindingDB (http://www.bindingdb.org/bind/index.jsp)
- Search for "succinate dehydrogenase iron-sulfur". Then click on "iron-sulfur subunit of complex ii".
- How many hits do you get?

What do I learn from the BindingDB hits?

- The relatively large number of compounds that bind to SDH suggests that the site is druggable
- Most of the results are from three studies, one that tested inhibition for a bacterium Septoria tritici and two for humans. The fact that many compounds bind to the human enzyme means that specificity could be an issue.
- It may be worth trying some of the compounds against *P. Aeruginosa*.

In the pharmaceutical industry, target selection is not only based on science, but also business

- How do pharmaceutical companies make money?
  - Costs
    - research and development is very expensive
    - manufacturing is relatively cheap
  - Income from drug sales
- In the current system, research and development(opposed to making old drugs) is incentivized by <u>patents</u>
  - must be novel, non-obvious, and useful. usually true for drugs.
  - provide monopoly on legal sales for a temporary period (in the U.S., 20 years from earliest filing)
  - need to be filed country-by-country
- In general, this system means that for a drug development campaign, the market must be
  - large enough
  - rich enough
  - not too competitive

Review questions

- What is necessary and what is desirable for a biological macromolecule to be a suitable target for structure-based drug design?
  - What is necessary/desirable about the physiological role?
  - What is necessary/desirable about the structural properties?
  - What are some business considerations that pharmaceutical companies use to decide whether to develop a drug for a particular target?
- How should we use large language models in identifying targets?
- What are some databases that you can use to help find and investigate a target?